Project Report : Uni-Network-v3

Campus Network Simulation & Cybersecurity Hardening

Prepared By: Madhur Guldagad

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1. Abstract

The "Uni-Network" project is a comprehensive simulation of a multi-building campus network topology designed using Cisco Packet Tracer. The primary objective of this project is to bridge theoretical networking concepts with practical implementation. The project is divided into two major phases: Phase 1 focuses on establishing core routing and switching infrastructure using protocols such as DHCP, VLANs, RIP, and NAT. Phase 2 introduces critical cybersecurity measures to harden the network against internal and external threats, implementing secure remote management (SSH), Layer 2 protection (Port Security), and traffic filtering (Extended Access Control Lists).

2. Introduction

Large-scale organizational networks, such as university campuses, require meticulous planning to ensure scalability, efficiency, and security. As networks grow, so do the attack surfaces available to malicious actors. This simulation models a realistic university environment with distinct administrative and academic subnets. By systematically configuring the network from the ground up, this project demonstrates the entire lifecycle of network engineering—from initial IP addressing and physical topology design to advanced security enforcement and traffic regulation.

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3. Phase 1: Core Network Infrastructure

The foundational phase of the Uni-Network topology ensures reliable and scalable data transmission across multiple physical locations.

3.1 Dynamic Host Configuration Protocol (DHCP)

To eliminate the administrative overhead of manually configuring IP addresses for hundreds of end devices, DHCP pools were configured on the core routers.

- **Implementation:** Separate IP pools were defined for the Student and Administration

networks. The routers dynamically lease IP addresses, subnet masks, default gateways, and DNS server information to end devices upon connection.

3.2 Virtual Local Area Networks (VLANs)

To manage broadcast domains and improve network performance, VLANs were deployed across the campus switches.

- **Implementation:** Network traffic was logically segregated based on departments. This ensures that a broadcast storm in the Student VLAN does not degrade the performance of the Administration VLAN, establishing a foundational layer of traffic isolation.

3.3 Routing Information Protocol (RIP)

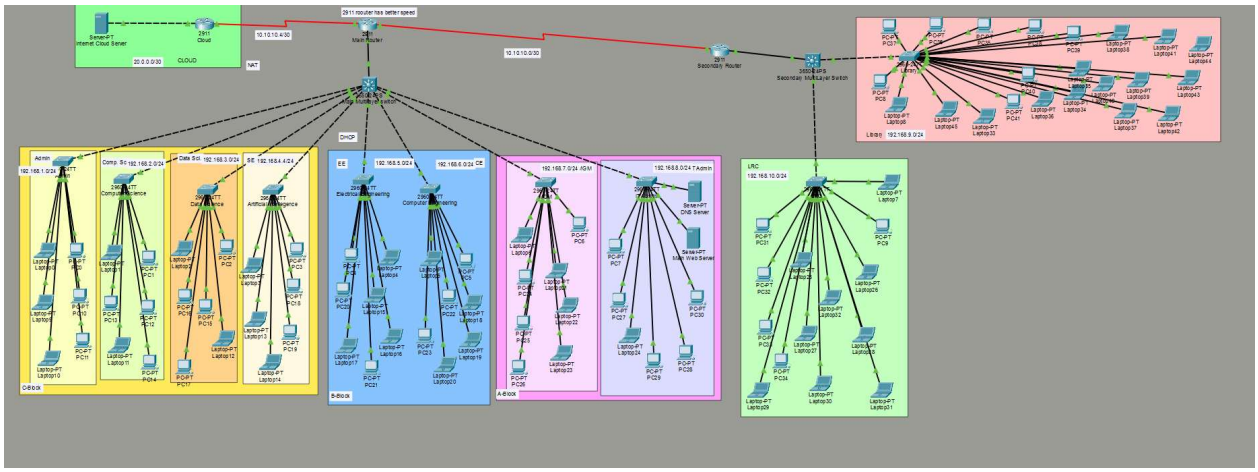
For inter-building communication, dynamic routing was necessary to allow different subnets to discover and communicate with one another.

- **Implementation:** RIP was enabled across all campus routers, allowing them to dynamically share routing tables and determine the most efficient paths for packet delivery across the topology.

3.4 Network Address Translation (NAT)

To allow internal devices residing on private IP addresses to access the external internet, NAT was configured at the network perimeter.

- **Implementation:** Internal private addresses were translated into a public IP address on the outbound router interface, successfully bridging the local campus network with external, simulated internet servers.



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4. Phase 2: Cybersecurity & Network Hardening

With functional connectivity established, the project focus shifted to securing the network against unauthorized access and data interception, aligning with modern cybersecurity standards.

4.1 Secure Device Management (SSH Integration)

Legacy management protocols like Telnet transmit data, including administrative passwords, in plaintext, making the network vulnerable to packet-sniffing attacks.

- **Implementation:** Telnet was deprecated globally across all routers and switches. RSA cryptographic keys (1024-bit) were generated using a localized domain name (uninetwork.com). The Virtual Teletype (VTY) lines were restricted to transport input ssh, ensuring all remote configuration sessions are heavily encrypted. A local user database was established to authenticate administrators securely.

4.2 Layer 2 Hardening (Port Security)

Open switch ports present a severe physical vulnerability, allowing rogue actors to connect unauthorized devices to the internal network.

- **Implementation:** Switch ports connected to end devices were hardcoded into access mode to prevent VLAN hopping. Port security was enabled with a maximum allowed device limit of 1. Using mac-address sticky, the switch dynamically learns and locks onto the MAC address of the first legitimate device connected. The violation protocol was set to shutdown, which immediately disables the interface if a foreign MAC address is detected.

4.3 Traffic Filtering (Extended Access Control Lists)

While VLANs provide basic isolation, routers will still route traffic between them by default. To enforce strict security policies, Extended Access Control Lists (ACLs) were deployed.

- **Implementation:** An internal firewall policy was established to protect sensitive data. An Extended ACL was written and applied to the router interface facing the student network.
    - **Deny Rule:** Explicitly drops any packets originating from the Student Subnet (192.168.10.0/24) destined for the Administration Server (192.168.20.50).
    - **Permit Rule:** Allows standard outbound traffic (such as internet browsing) to pass unhindered.
    - **Direction:** Applied inbound (in) to stop unauthorized traffic at the closest possible point to the source.

```
Router(config)#hostname R1
R1(config)#ip domain-name uninetwork.com
R1(config)#crypto key generate rsa
The name for the keys will be: R1.uninetwork.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#username admin secret password123
*Mar 1 0:35:22.10: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#transport input ssh
R1(config-line)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
R1#show access-lists
Standard IP access list 1
    10 permit 192.168.0.0 0.0.255.255
    20 permit 192.168.0.0 0.0.15.255

R1#
```

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5. Testing and Verification

To ensure the integrity of the network and the active status of the security protocols, rigorous testing was conducted.

- **DHCP & Connectivity:** End devices successfully requested and received IP leases. Ping tests between buildings validated the RIP routing tables.
- **SSH Verification:** Attempts to connect via Telnet were actively refused by the network devices, while SSH connections successfully prompted for encrypted credential authentication.
- **Port Security Verification:** Simulating a rogue laptop connection resulted in the target switch port immediately transitioning to a down state, triggering a red link indicator in the topology.

```
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa 0/1
Switch(config-if)#swi mode acc
Switch(config-if)#swi port-secu
Switch(config-if)#swi port-secu max 1
Switch(config-if)#swi port-secu mac-address sticky
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show port-se
Switch#show port-security int fa 0/1
Port Security              : Enabled
Port Status                : Secure-shutdown
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 0
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : 0000.0000.0000:0
Security Violation Count   : 0

Switch#
```

- **ACL Verification:** ICMP ping requests initiated from a Student PC to the Administration Server returned an immediate "Destination host unreachable" response. Subsequent execution of the show access-lists command on the core router confirmed positive match counts for the dropped packets, proving the firewall rules were actively filtering traffic.

```
Command Prompt                                                    X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.50.20

Pinging 192.168.50.20 with 32 bytes of data:

Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.

Ping statistics for 192.168.50.20:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

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6. Conclusion & Future Scope

The Uni-Network project successfully demonstrates the evolution of a campus network from a basic, vulnerable state to a highly functional, hardened architecture. By implementing dynamic addressing, segmented routing, and rigorous security protocols, the simulation reflects industry-standard networking practices.

Future Scope:

In subsequent iterations of this project, the architecture will be expanded to include:

1. **DHCP Snooping:** To prevent rogue DHCP server attacks from poisoning the network's IP allocation.
2. **Centralized Syslog Server:** To aggregate, monitor, and analyze security alerts generated by port security violations and ACL drops in real-time.
3. **Topology Expansion:** Integrating additional academic buildings and specialized IoT (Internet of Things) laboratory subnets.